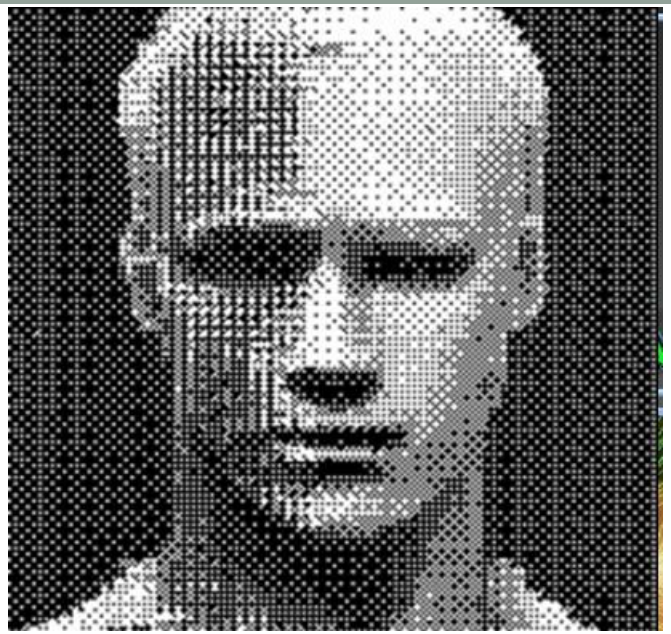


DIGITAL IMAGE PROCESSING

Dr. Manar Donia

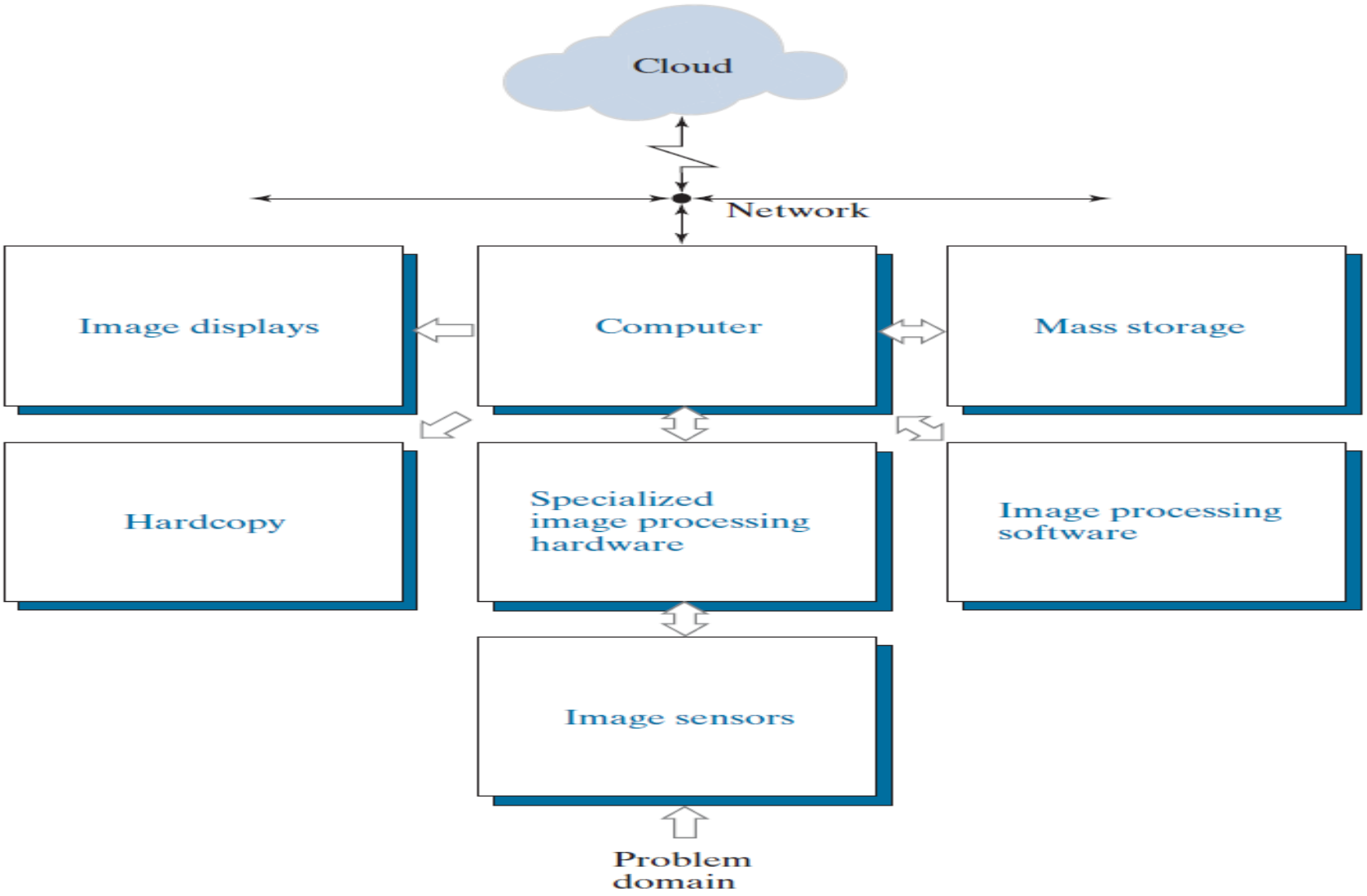
CMPn332 - CMPN332 & CMP 432

What is the objective of image processing?

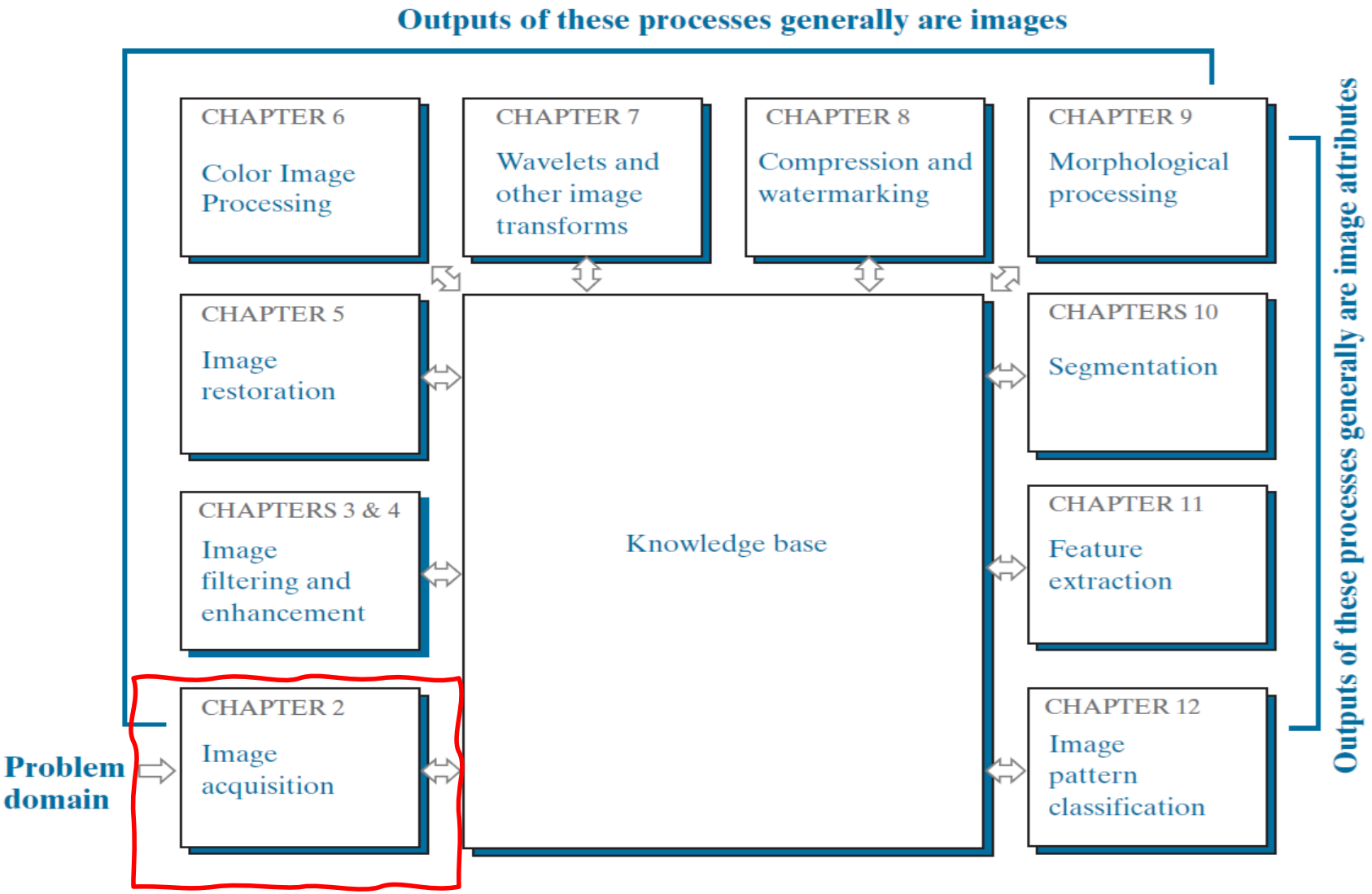
The ultimate goal of digital image processing is to build systems that can automatically analyze, interpret, enhance, and manipulate digital images such systems are:

- **Medical Imaging System**→ build a system that can detect tumors in medical scans.
- **Security and Surveillance System**→ Image processing is crucial for real-time monitoring and detection of suspicious activities in security systems.
- **Biometric Security System**→ Image processing algorithms are used to capture, enhance, and analyze fingerprints for identification purposes.
- **License Plate Recognition System**→ Recognizing vehicle license plates in traffic enforcement.

Basic components of any DIP system

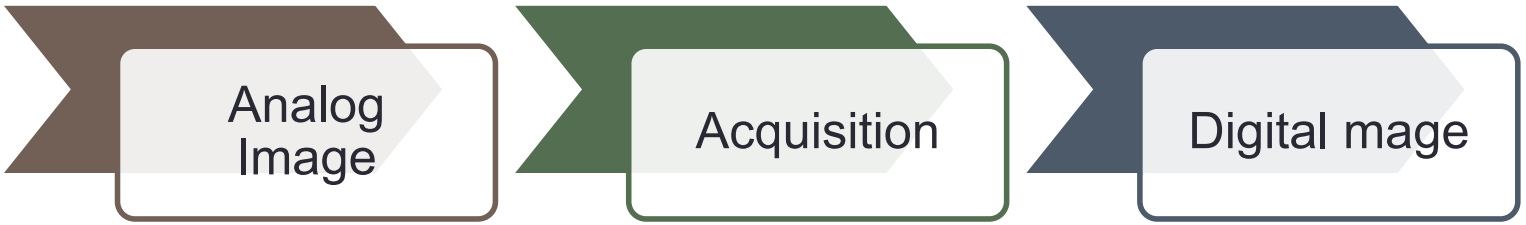


Fundamental steps in digital image processing

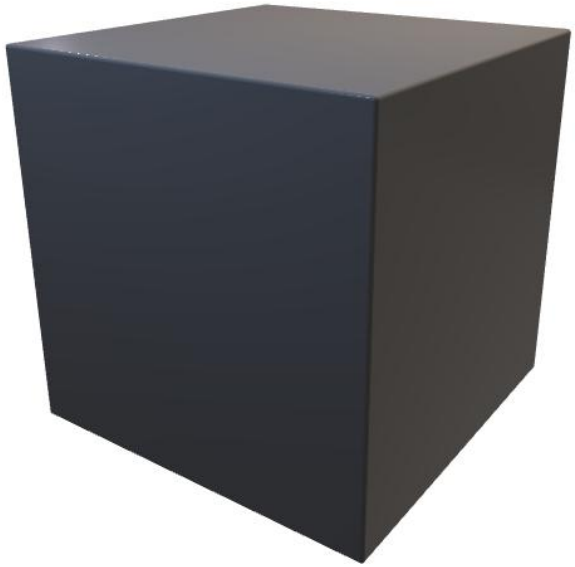


Fundamental steps in digital image processing

- Image Acquisition:
  - It's the first step in any image processing system that captures visual information from the real world and converts it into a digital image that computers can process..



What's an Image?

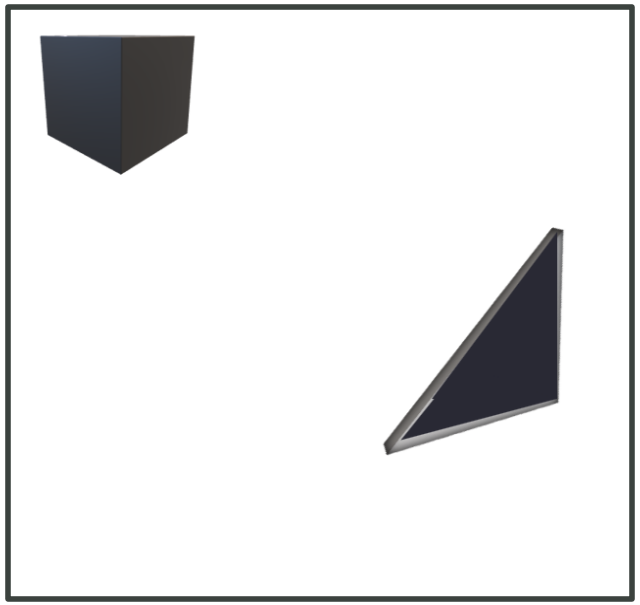




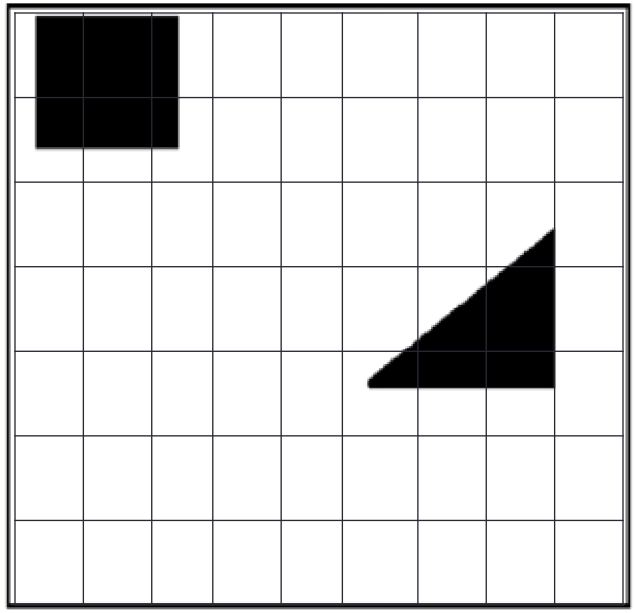
- Image is a 2D projection of 3D object.
- So, image can be defined as 2D function F(x, y)
  - where x, y represent spatial coordinates
  - and the amplitude of F at any point is called the intensity at that point.

Image Types

- Analog Image and Digital Image

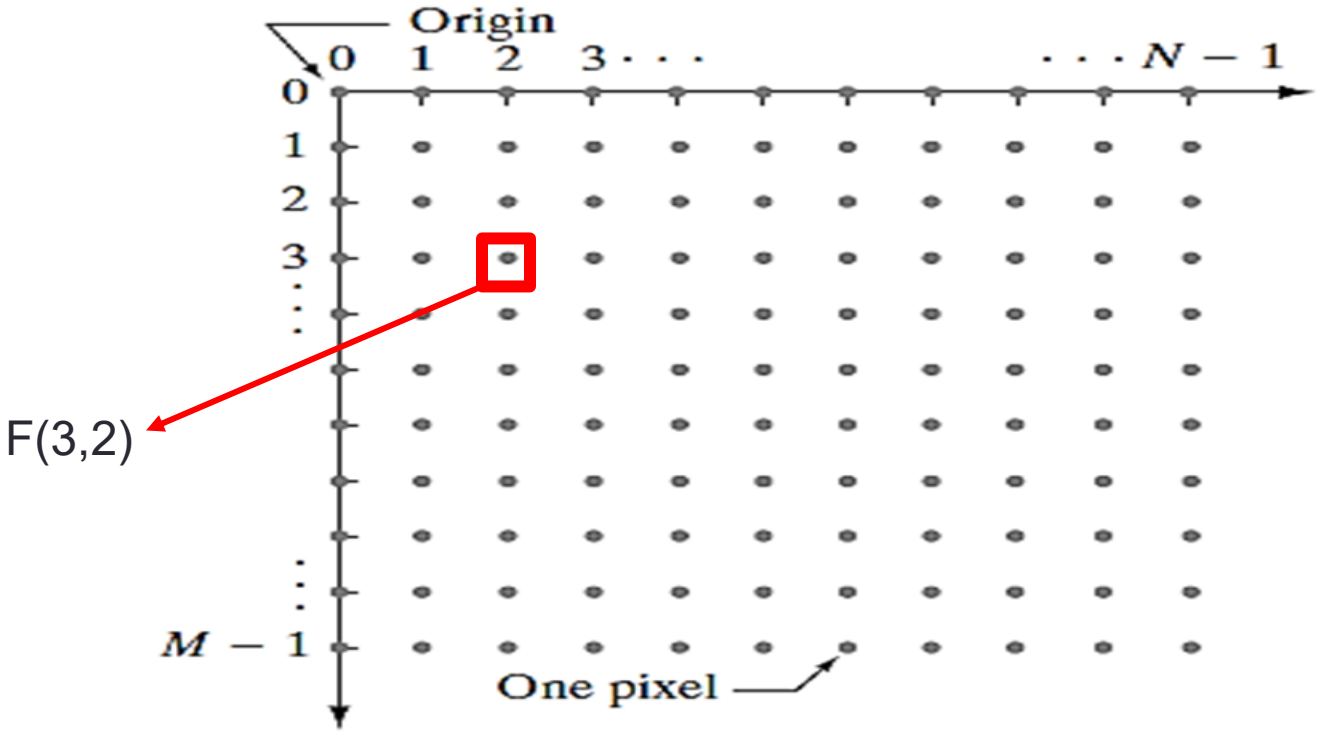


Analog Image→ continuous variation in tones



Digital Image→ finite set of digital values called pixels

Digital Image representation



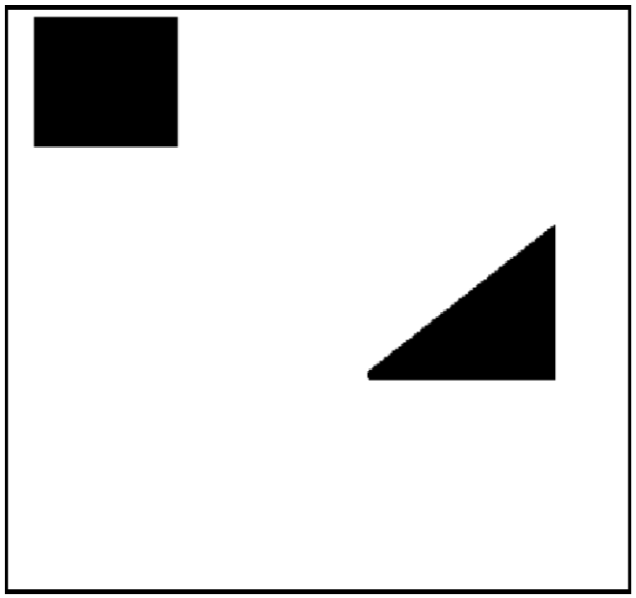
To refer to a specific pixel within the image matrix, we define its coordinate at row and column.

Digital Image representation

To convert Analog image into Digital we do:

Sampling: Digitizing the co-ordinate value.

Quantization: Digitizing the amplitude value.



25 x 25 image matrix

```
Image Matrix =
[255, 210, 202, 205, 204, 201, 203, 206, 204, 219, 255, 255, 255, 255, 253, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 253,  42,   2,   1,   0,   4,   1,   0,   1,  67, 252, 255, 255, 254, 253, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 254,  49,   1,   0,   3,   0,   0,   2,   2,  70, 255, 255, 249, 255, 255, 252, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255,  51,   0,   0,   0,   3,   0,   0,   3,  72, 255, 252, 255, 255, 254, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 254,  47,   0,   6,   0,   0,   4,   0,   0,  75, 255, 254, 255, 253, 253, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255,  54,   0,   0,   2,   0,   3,   0,   0,  75, 249, 255, 255, 255, 255, 250, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 254,  40,   3,   1,   0,   3,   0,   0,   0,  65, 255, 253, 253, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255,  85,  39,  45,  39,  45,  43,  44,  43, 106, 252, 255, 254, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 253, 255, 255, 255, 254, 255, 139, 209;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 254, 254, 255, 254, 133,   0, 210;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 251, 255, 251, 255, 120,   0,   1, 213;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 255, 112,   1,   1,   0, 215;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 253, 101,   0,   0,   0,   0, 214;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 253, 255,  96,   0,   2,   3,   1,   0, 214;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 254, 253,  83,   0,   2,   0,   0,   4,   0, 215;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 254, 253,  76,   0,   1,   0,   1,   0,   0,   2, 214;
 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 255, 255, 255, 254, 227,  45,   0,   0,   0,   0,   0,   0,   0,   0, 211;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 255, 255, 252, 206, 143, 155, 155, 155, 155, 155, 155, 155, 155, 240;
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 255, 255, 255, 255, 255, 255, 255, 255, 252, 255, 255, 255, 252, 252, 255, 252, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 252, 255, 253, 254, 255, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 251, 255, 255, 253, 255, 252, 255, 255, 255, 255, 255, 255, 255, 255, 254;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 249, 254, 255, 252, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 255, 254, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255]
```

Digital Image representation

Image →8 x 8 pixels



F =

```
[255, 255, 252, 255, 255, 255, 255, 253;
 254, 255, 253, 254, 255, 253, 255, 255;
 255, 255, 255, 255, 255, 255, 253, 252;
 255, 250, 255, 254, 165, 145, 149, 204;
 251, 255, 254, 253,  26,   4,   0, 120;
 255, 255, 255, 254,  31,   0,   0, 120;
 252, 255, 252, 255,  87,  58,  56, 159;
 255, 253, 255, 255, 254, 255, 255, 254]
```

The amplitude values are defined by bits (pixel depth)

Pixel depth(k) (k=number of bits per pixel)	Maximum gray levels ($L = 2^k$)	gray level interval 0→(L-1)
1	2	0,1
2	4	0→3
3	8	0→7
8	256	0→255

Digital Image storage and transmission

- When considering image storage and transmission requirements, you must consider how much space an image occupies.

- For M x N digital image and k bits (pixel depth)
  - Number of bits required to store the image is

$$M*N*k$$

- For M x N digital image and L Maximum gray levels
  - Number of bits required to store the image is

$$M*N*\text{Log}_2(L)$$

Digital Image color models

RGB



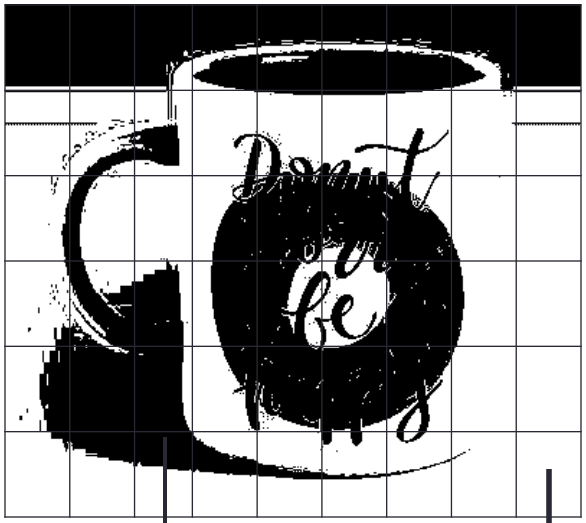
131,173,39

Grayscale



127

B&W



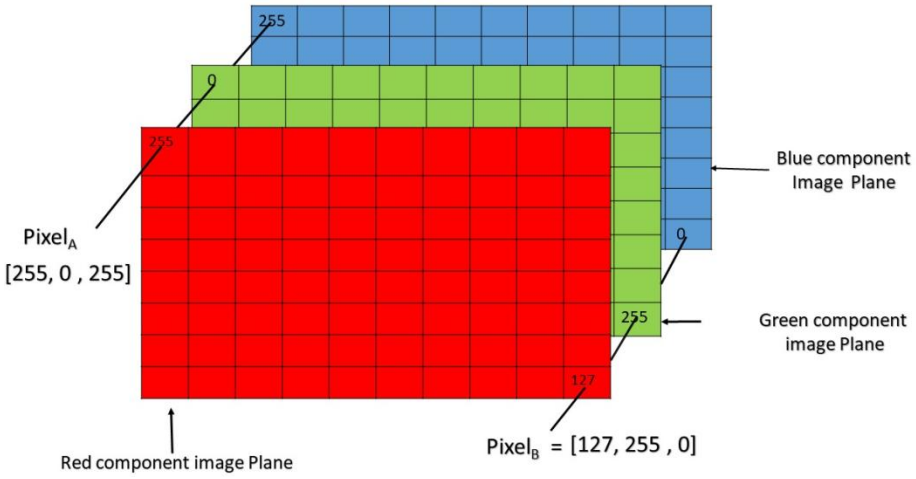
0

255

Digital image is composed of a finite number of pixels **"picture elements"**, each of which has a particular location and value

Digital Image color models

RGB

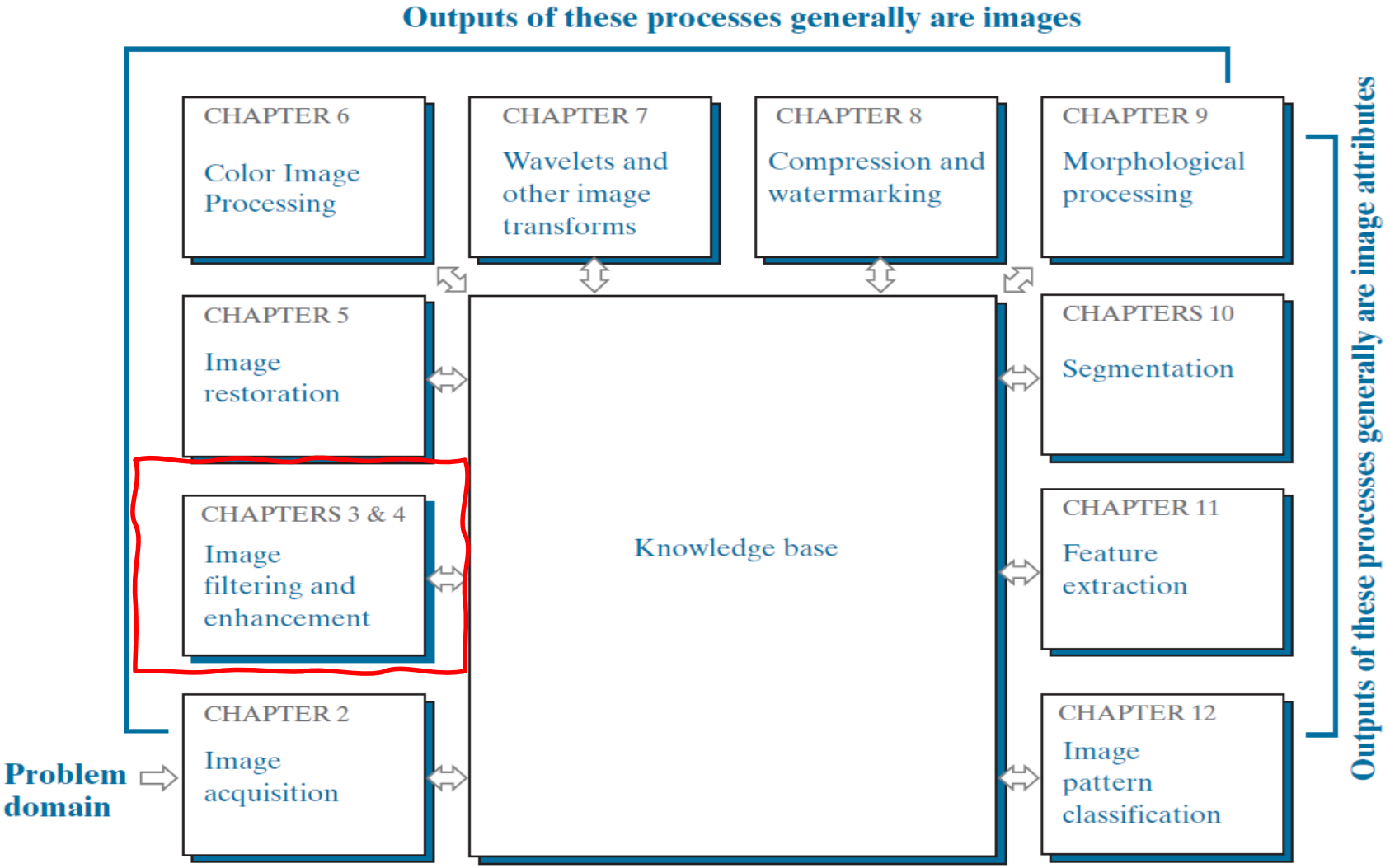


Grayscale

$$\begin{bmatrix} 52 & 55 & 61 & 66 & 70 & 61 & 64 & 73 \\ 63 & 59 & 55 & 90 & 109 & 85 & 69 & 72 \\ 62 & 59 & 68 & 113 & 144 & 104 & 66 & 73 \\ 63 & 58 & 71 & 122 & 154 & 106 & 70 & 69 \\ 67 & 61 & 68 & 104 & 126 & 88 & 68 & 70 \\ 79 & 65 & 60 & 70 & 77 & 68 & 58 & 75 \\ 85 & 71 & 64 & 59 & 55 & 61 & 65 & 83 \\ 87 & 79 & 69 & 68 & 65 & 76 & 78 & 94 \end{bmatrix}$$

In our work we mostly work on grayscale system as RGB image has 3D while grayscale has just one, so, everything tend to be more costly in RGB system

Fundamental steps in digital image processing



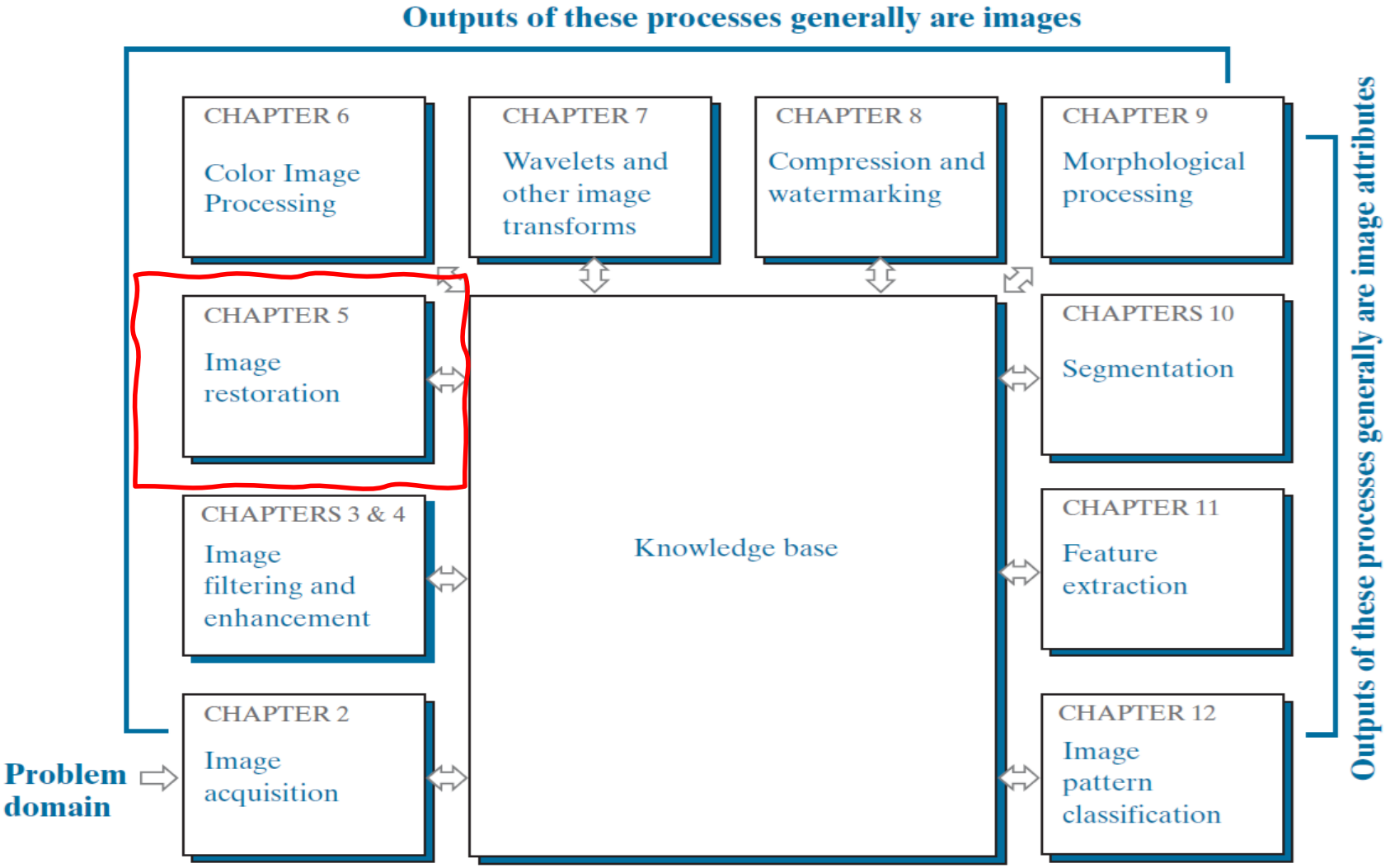
Fundamental steps in digital image processing

- Image enhancement:
  - It is the process of improving the visual quality of an image, making it more suitable for further automated analysis



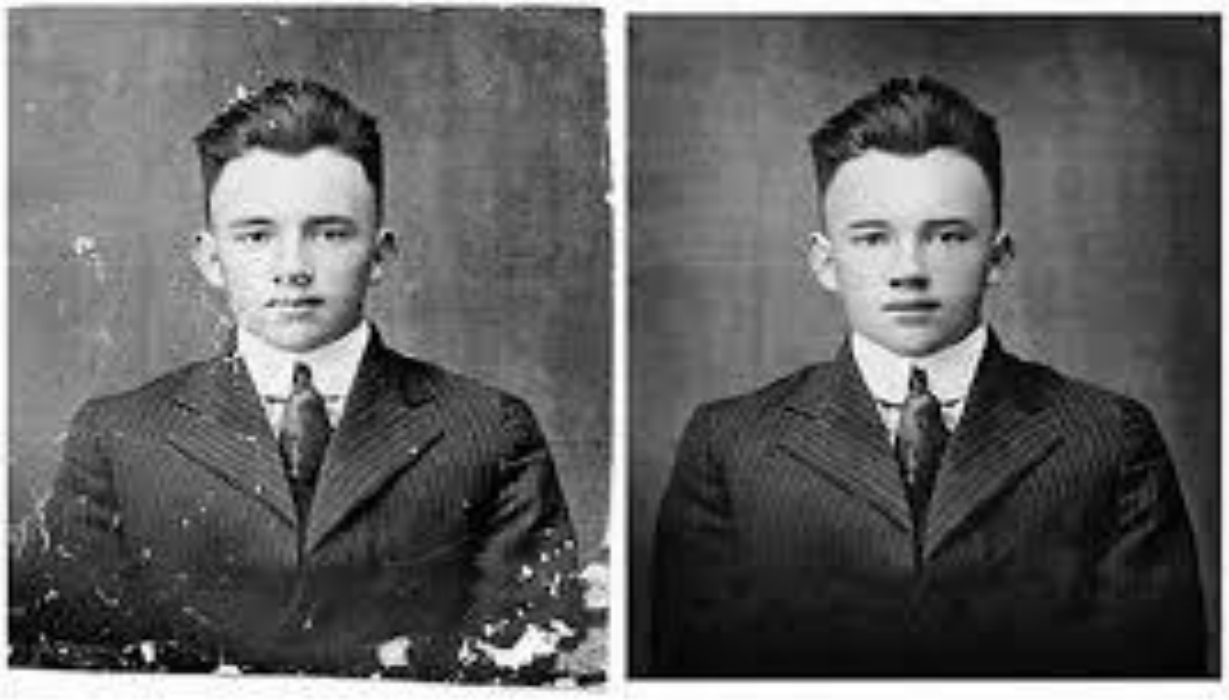


Fundamental steps in digital image processing

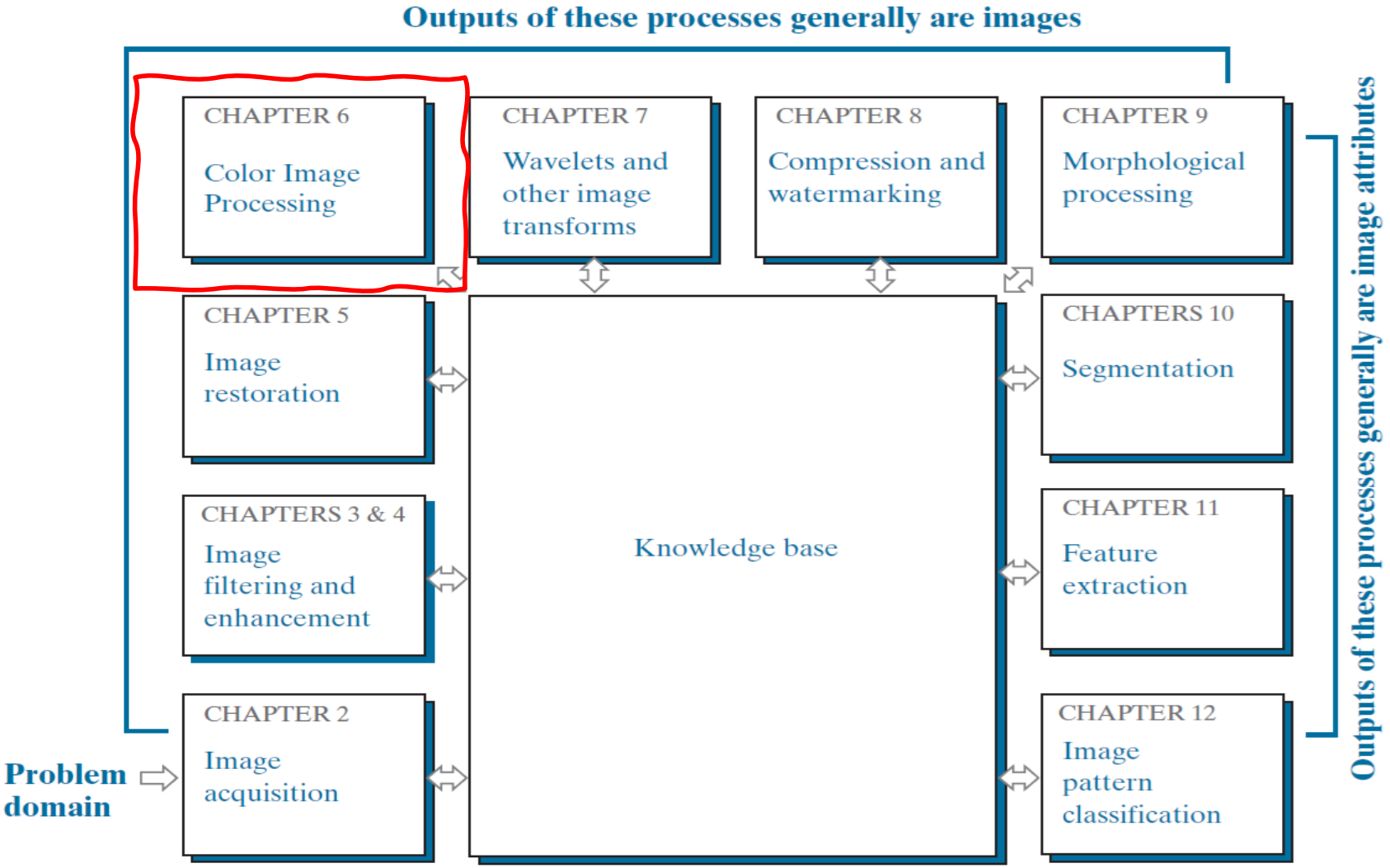


Fundamental steps in digital image processing

- Image restoration:
  - It is the process of recovering or reconstructing an image that has been degraded due to various factors such as noise, blur, or loss of information.

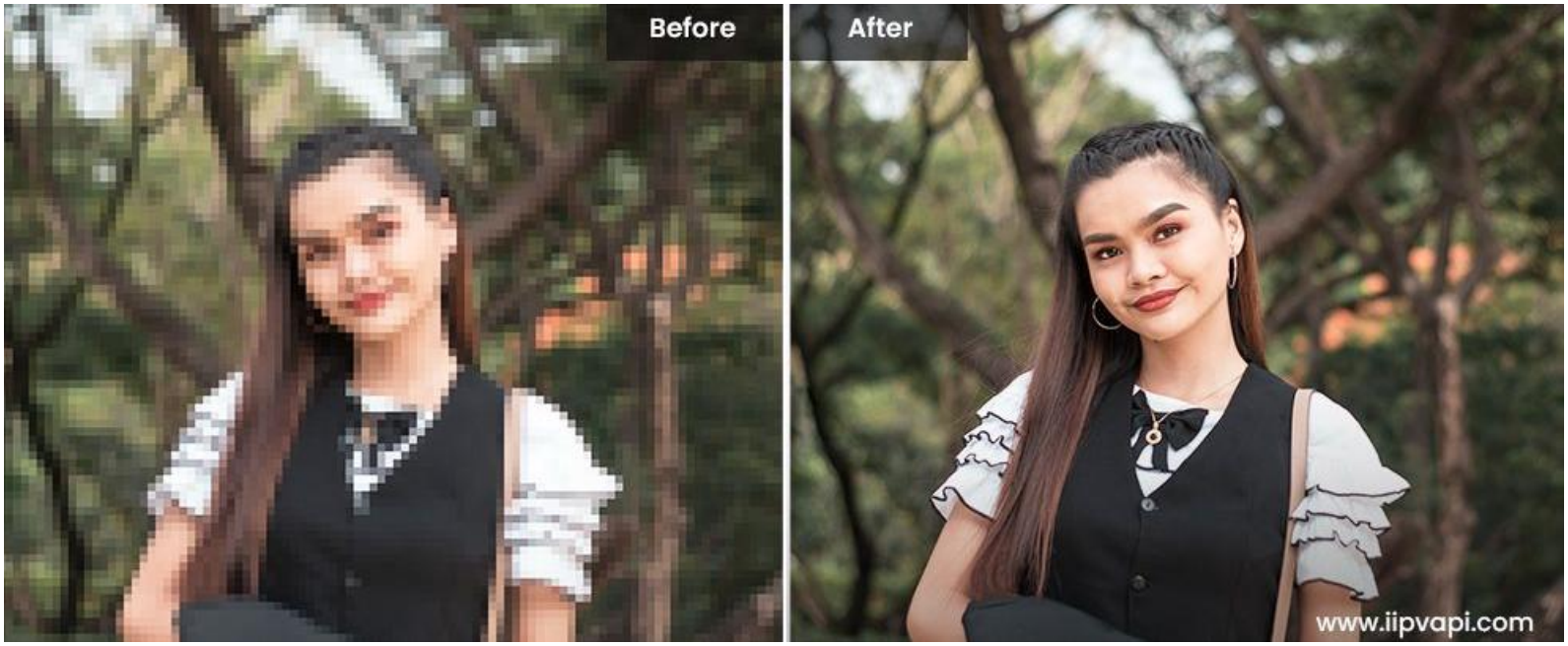


Fundamental steps in digital image processing

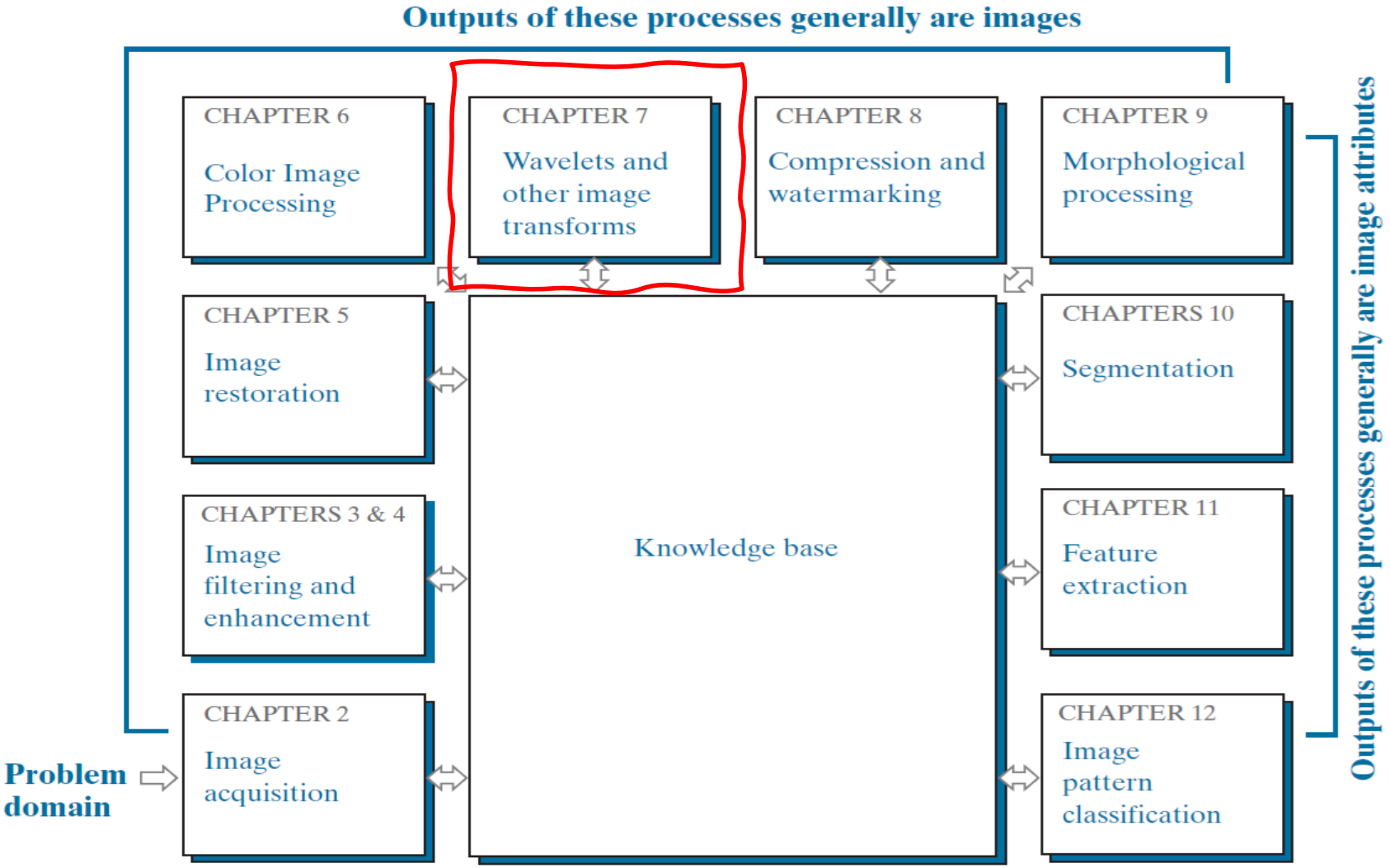


Fundamental steps in digital image processing

- Color Image Processing:
  - Color images consist of multiple color channels, It involves the manipulation and analysis of color images, which contain more complex information than grayscale images

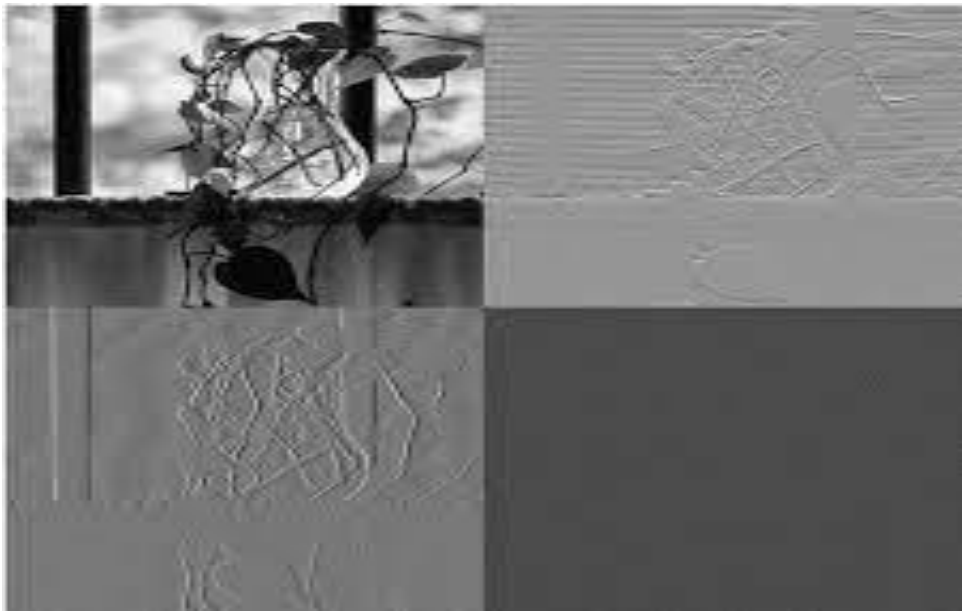


Fundamental steps in digital image processing

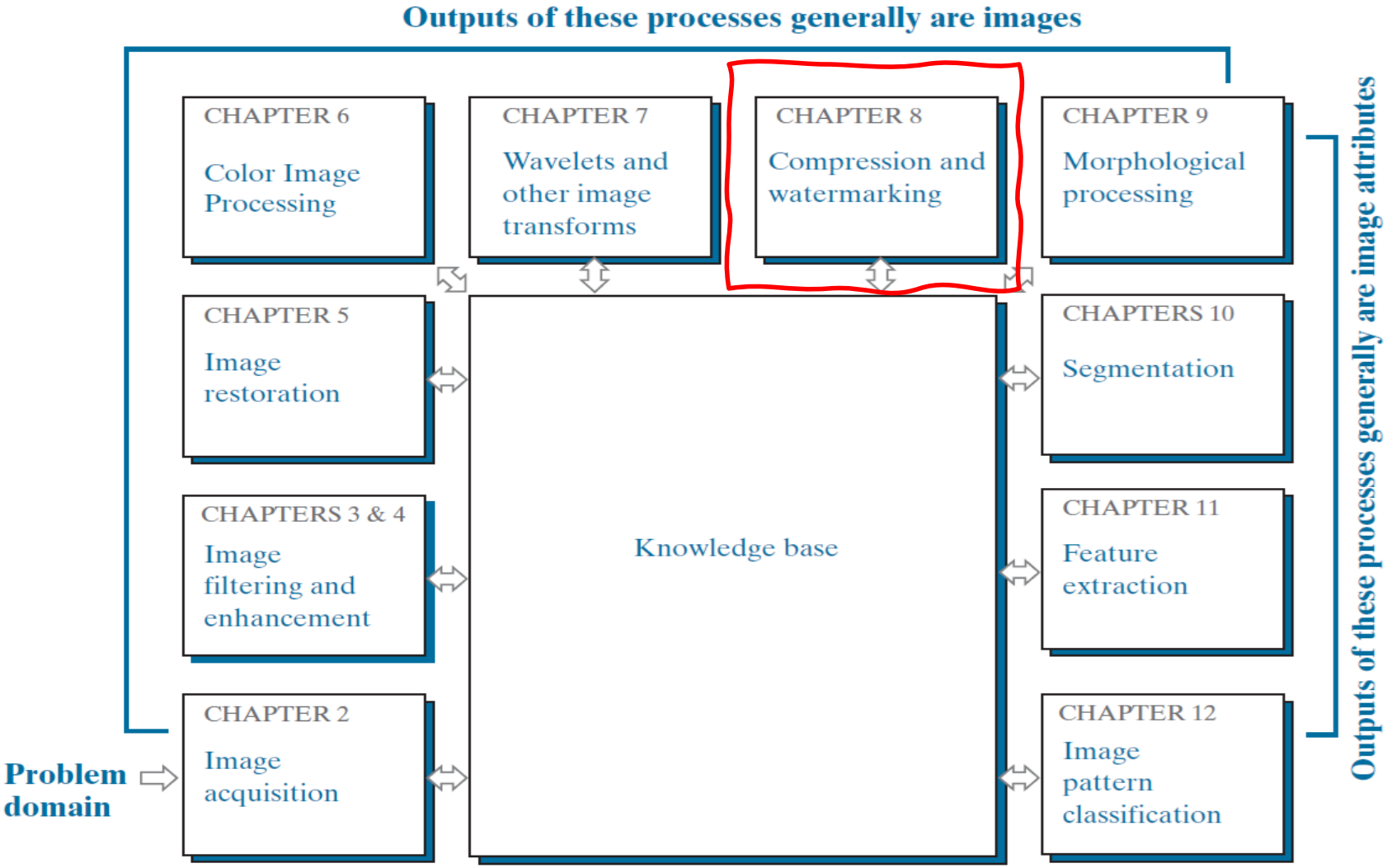


Fundamental steps in digital image processing

- Wavelets and multiresolution processing:
  - It involves transforming an image into a set of wavelet coefficients, which capture both the frequency and spatial (or temporal) information at multiple scales.



Fundamental steps in digital image processing

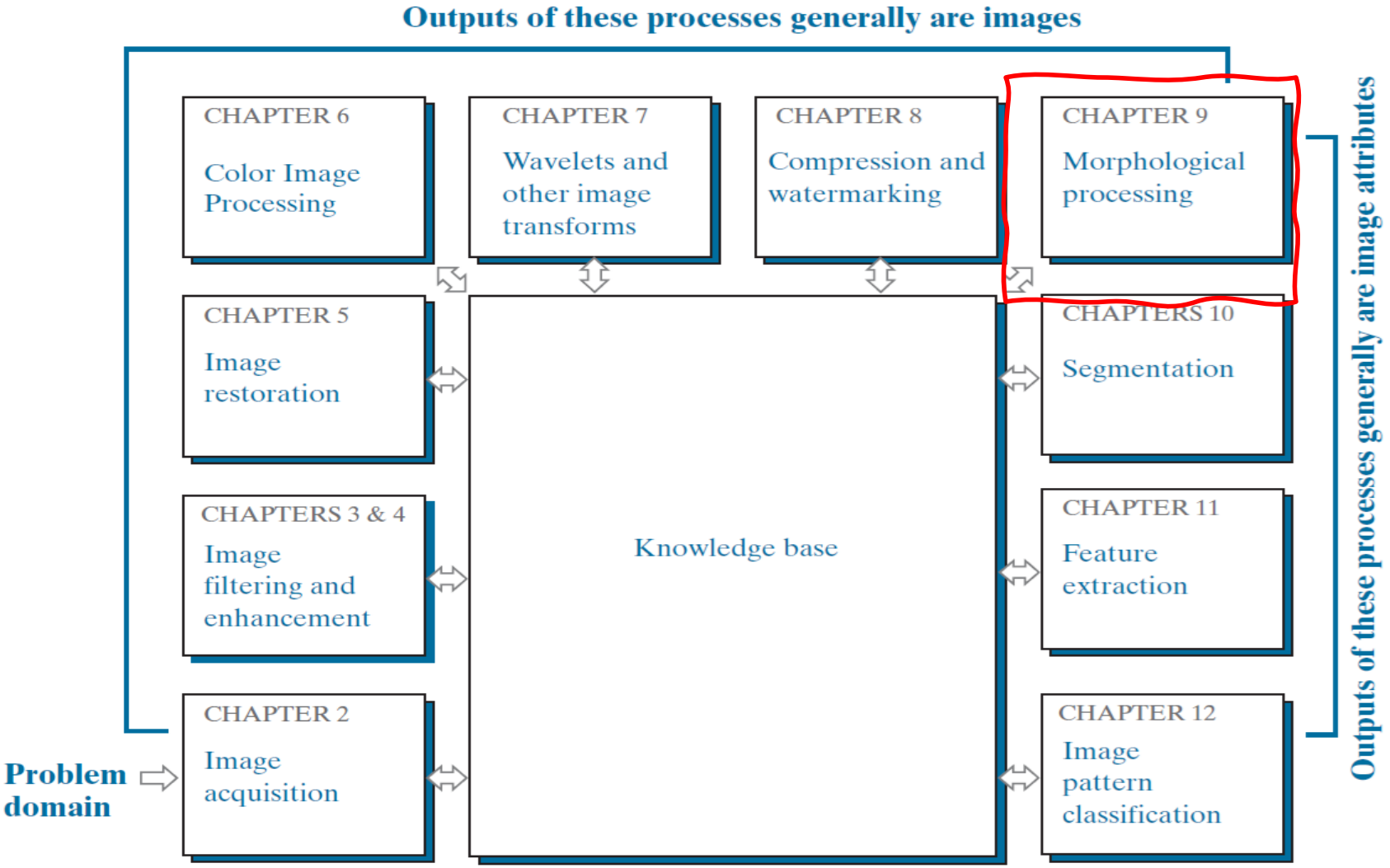


Fundamental steps in digital image processing

- Image compression:
  - It is the process of reducing the size of a digital image file while maintaining an acceptable level of image quality.

- Image Compression types
  - Lossless (Exact)→ Original = Decompressed
  - Lossy →Original ≅ Decompressed

Fundamental steps in digital image processing

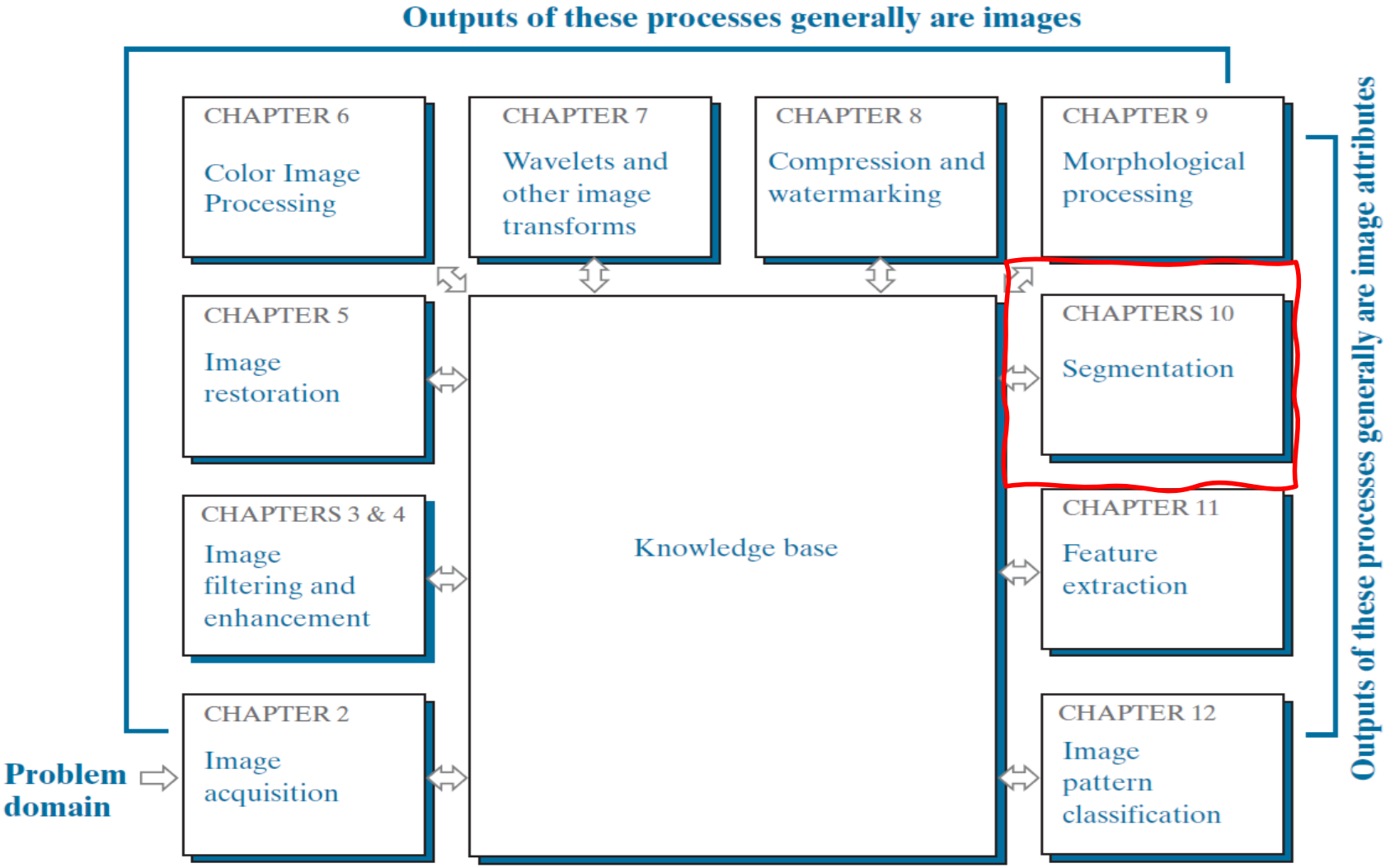


Fundamental steps in digital image processing

- Morphological processing:
  - Morphological processing is a tool for shape-based image analysis. it is used to extract the boundaries of objects in an image.



Fundamental steps in digital image processing

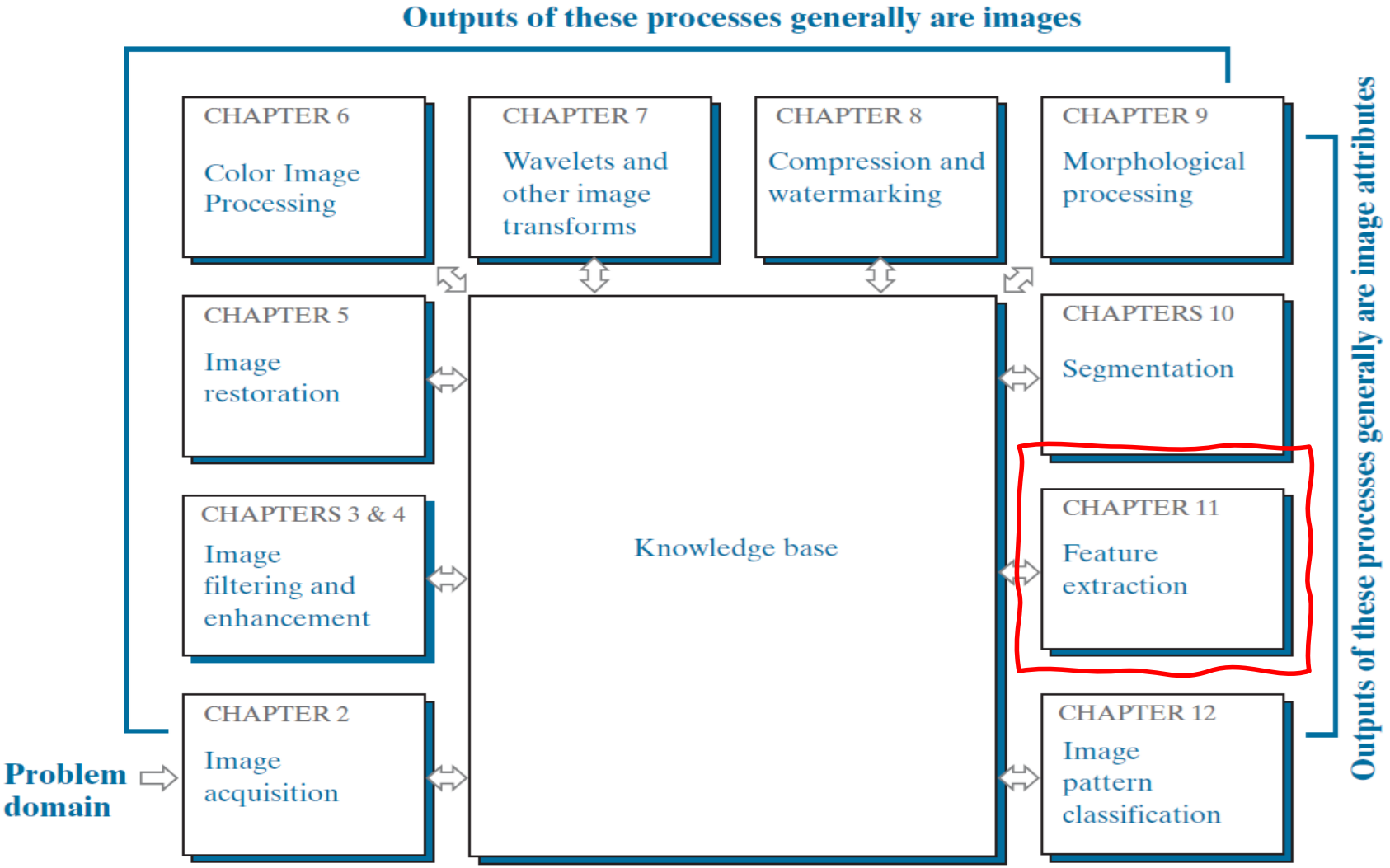


Fundamental steps in digital image processing

- Image segmentation:
  - It involves dividing an image into distinct regions or segments to simplify its representation and make it easier to analyze.



Fundamental steps in digital image processing

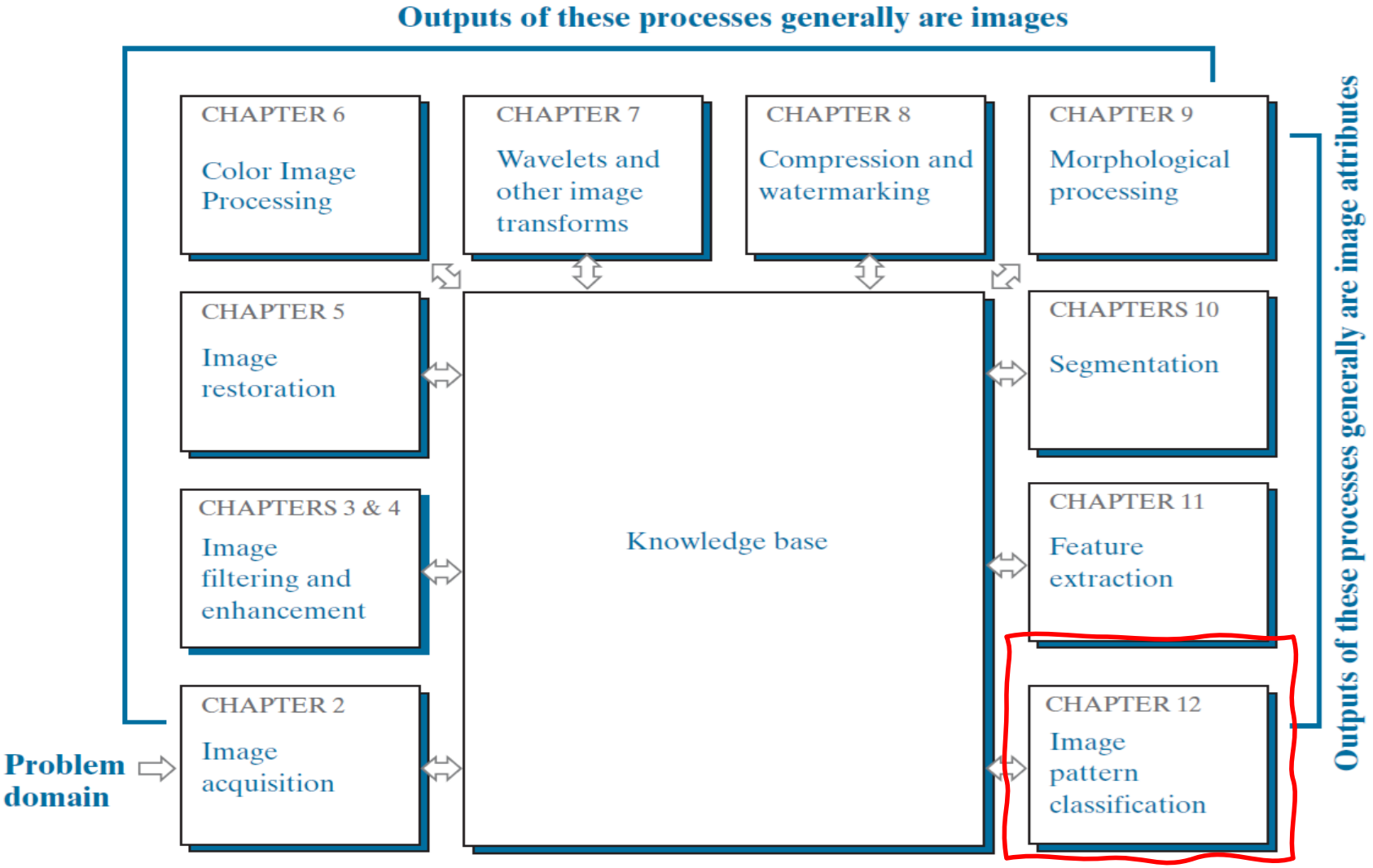


Fundamental steps in digital image processing

- **Feature extraction** consists of feature detection and feature description.
  - Feature detection refers to finding the features in an image, region, or boundary.
  - Feature description assigns quantitative attributes to the detected features.

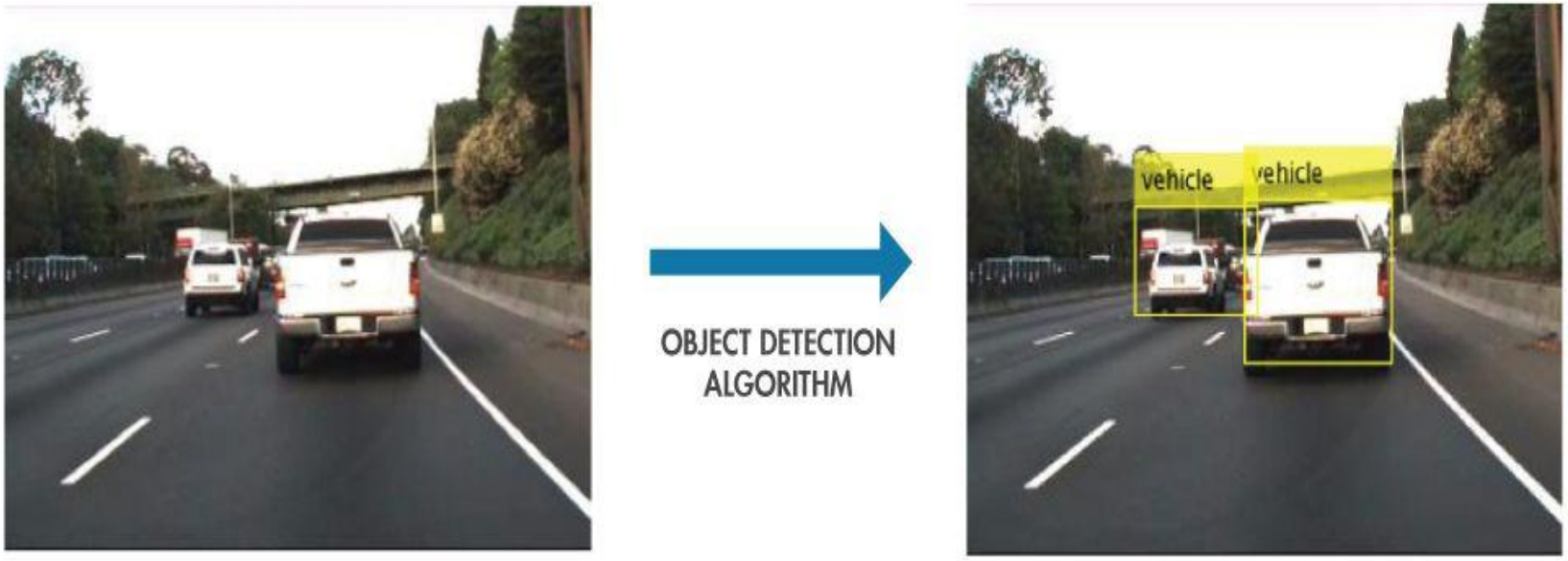
  - For example, we might detect corners in a region boundary, and describe those corners by their orientation and location, both of which are quantitative attributes.

Fundamental steps in digital image processing



Fundamental steps in digital image processing

- Object detection:
  - It's a fundamental task in computer vision that involves identifying and classifying objects within an image or video stream.




Thank you!